

Efficient Taxonomy Tagging for Educational Content

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Student Declaration

We hereby declare that the work presented in the report entitled "**Efficient Taxonomy Tagging for Educational Content**" submitted by us for the partial fulfilment of the requirements for the degree of at Indraprastha Institute of Information Technology, Delhi, is an authentic record of our work carried out under the supervision of **Prof. Vikram Goyal** and **Prof. Mukesh Mohania**, and with regular academic guidance from **Prof. Venkatesh V, Assistant Professor, Stockholm University**. Due acknowledgements have been given in the report for all material used. This work has not been submitted elsewhere for the award of any other degree.

Harsh Rajput

Place & Date: April 21, 2026

Certificate

This is to certify that the above statement made by the candidates is correct to the best of my knowledge.

Prof. Vikram Goyal
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Abstract

Educational platforms organize questions using hierarchical taxonomies (Subject–Chapter–Topic), but automatically tagging new questions remains challenging due to label imbalance, semantic mismatch, and noisy weak supervision. We adapt a retrieval-based tagging approach, aligning question embeddings with hierarchical NCERT labels to generate initial weak labels at scale. However, LLM-generated confidence scores are often miscalibrated and cannot be directly trusted. To obtain reliable training data with minimal expert effort, we combine a small human-labeled seed set with prediction-powered inference (PPI) and introduce a local variant, Local PPI (L-PPI), which estimates instance-specific bias and uncertainty using nearby human-labeled examples in embedding space. L-PPI produces calibrated scores and confidence intervals for every question–label pair, enabling statistically grounded filtering of high-confidence annotations. This pipeline yields a cleaner dataset for iterative tagger retraining and improves robustness in hierarchical taxonomy tagging for educational content.

Keywords: Hierarchical Taxonomy Tagging, Dense Retrieval, Contrastive Representation Learning, Large Language Models, Weak Supervision, Prediction-Powered Inference (PPI), Local PPI (L-PPI), Calibration, Educational Data Mining, NCERT Taxonomy.

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Chapter 1

Introduction

1.1 Background

The rapid growth of digital educational content has enabled large-scale personalized learning, intelligent tutoring systems, and adaptive assessment platforms. In K–12 and competitive exam ecosystems (e.g., JEE), vast repositories of practice questions and learning materials are continuously generated. While this abundance improves accessibility, it also introduces significant challenges in organizing, retrieving, and delivering relevant content effectively.

Educational platforms address this challenge using **hierarchical taxonomies**, typically structured as *Subject* $\rightarrow$ *Chapter* $\rightarrow$ *Topic*. Such taxonomies enable fine-grained categorization, targeted learning pathways, and personalized recommendations. Consequently, accurate taxonomy classification of questions is a fundamental component of modern educational systems.

Existing taxonomy classification models, such as *TagRec* and its variants, attempt to automate this process using retrieval-based and learning-based approaches. However, a major limitation in real-world deployment is that **educational taxonomies are dynamic**. Syllabi evolve frequently, with topics being added, removed, or restructured, and new question patterns emerging over time. This continuous evolution makes it difficult to maintain up-to-date labeled datasets required for training and adapting these models.

1.2 Motivation

The primary challenge in taxonomy classification is the **high cost of annotation** required to map questions to fine-grained hierarchical labels. This challenge is exacerbated in dynamic settings where taxonomy structures change frequently, rendering previously annotated datasets partially obsolete. As a result, repeatedly constructing large-scale labeled datasets is both expensive and impractical.

A natural alternative is to leverage existing taxonomy classification models to generate labels for new, unlabeled question corpora in a zero-shot or weakly supervised manner. However, such automatically generated labels are inherently noisy and require quality assessment before being used for

training.

Recent advances in Large Language Models (LLMs) provide a promising mechanism for evaluating the quality of predicted labels by scoring the alignment between a question and its assigned taxonomy label. Despite their strong reasoning capabilities, these scores are often **miscalibrated**, exhibiting biases, inconsistencies across domains, and sensitivity to prompt design. Consequently, relying directly on such scores can lead to the selection of low-quality labels, negatively impacting downstream model performance.

This motivates the need for methods that can **reliably assess and refine automatically generated labels with minimal human supervision**, enabling scalable dataset creation for evolving taxonomies.

1.3 Problem Statement

Given a hierarchical taxonomy T and a large corpus of unlabeled questions U , along with an initial taxonomy classification model that produces candidate labels, the goal is to construct a high-quality labeled dataset for training improved classification models while minimizing human annotation effort.

Formally, the objective is to learn a function

$$f : Q \rightarrow T$$

by leveraging weak supervision signals derived from model predictions and auxiliary scoring mechanisms.

This problem involves several key challenges:

- **Dynamic Taxonomy Evolution:** Frequent changes in taxonomy structure require continuous adaptation of labeled datasets and models
- **Noisy Pseudo-Labels:** Labels generated by initial models in zero-shot settings may be inaccurate or inconsistent
- **Unreliable Scoring Signals:** Automated scoring mechanisms (e.g., LLM-based evaluators) may produce miscalibrated confidence estimates
- **Limited Supervision:** Only a small fraction of data can be feasibly annotated by humans due to cost constraints
- **Data Selection:** Identifying high-quality labeled instances from noisy predictions is critical for effective training
- **Scalability:** The approach must scale to large datasets and extensive label spaces without excessive computational overhead

The central problem is therefore to develop a framework that can **transform noisy, weakly supervised signals into reliable training data**, enabling accurate and scalable taxonomy classification under evolving taxonomies with minimal human annotation.

1.4 Research Contributions

This thesis makes the following key contributions toward scalable and reliable taxonomy classification under evolving educational taxonomies:

1.4.1 Instance-Level Calibration of Weak Supervision Signals

We propose a novel instance-level calibration framework inspired by Prediction-Powered Inference (PPI) [2], adapted to refine noisy supervision signals obtained from automated evaluators. Unlike traditional PPI methods designed for population-level estimation, our approach focuses on **calibrating confidence scores at the individual instance level**.

This enables:

- **Bias correction of automated scores** using a small human-annotated calibration set
- **Improved reliability of LLM-based evaluations** by addressing systematic miscalibration
- **Instance-specific confidence estimation** for downstream data selection
- **Robust filtering of high-quality pseudo-labels** from noisy predictions

1.4.2 Scalable Pipeline for Dataset Construction under Evolving Taxonomies

We develop an end-to-end framework for constructing high-quality labeled datasets from large unlabeled question corpora with minimal human supervision. The proposed pipeline integrates:

- **Weak Supervision:** Using an initial taxonomy classification model (e.g., TagRec) to generate candidate labels for unlabeled questions
- **LLM-based Evaluation:** Scoring the alignment between questions and predicted taxonomy labels
- **Calibration with Limited Supervision:** Leveraging a small human-annotated subset to correct scoring biases
- **Quality-based Filtering:** Selecting high-confidence question-label pairs to construct a reliable training dataset
- **Model Refinement:** Training improved taxonomy classification models on the filtered dataset

We demonstrate that models trained on such **calibrated and filtered pseudo-labeled data** can achieve performance comparable to models trained on fully human-annotated datasets of the same scale, thereby significantly reducing human annotation costs.

1.5 Thesis Objectives

The primary objectives of this research are:

1. To develop an instance-level calibration method for refining noisy evaluation signals using a small human-annotated dataset
2. To design a scalable pipeline for constructing high-quality labeled datasets from large unlabeled question corpora
3. To enable efficient adaptation of taxonomy classification models to evolving taxonomies with minimal annotation effort
4. To empirically demonstrate that calibrated and filtered pseudo-labeled data can match the performance of fully human-labeled datasets
5. To establish a general framework for weakly supervised learning in large-scale hierarchical classification settings

The research presented in this thesis represents a significant step toward practical and scalable educational content organization, with potential applications in intelligent tutoring systems, adaptive learning platforms, and educational resource management. By addressing the fundamental challenge of annotation efficiency, this work enables more widespread adoption of detailed taxonomic organization in educational technology.

Chapter 2

Literature Review

The problem of organizing large-scale educational content into structured taxonomies has been widely studied across information retrieval and natural language processing. Prior work primarily focuses on automated taxonomy classification, leveraging representation learning and hierarchical modeling techniques. More recently, the emergence of Large Language Models (LLMs) has introduced new paradigms for evaluation and weak supervision, while statistical frameworks such as Prediction-Powered Inference (PPI) have enabled principled calibration of noisy signals. In this section, we review relevant literature across these directions.

2.1 Taxonomy Classification Models

Early approaches to educational taxonomy classification primarily rely on supervised learning and retrieval-based methods. [8] propose TagRec, which formulates taxonomy classification as a dense retrieval problem by aligning question embeddings with taxonomy label representations. This approach avoids explicit multi-class classification over large label spaces and demonstrates improved scalability for hierarchical taxonomies. Building upon this, [9] introduce TagRec++, incorporating hierarchical label-aware attention mechanisms and improved negative sampling strategies to better capture relationships across taxonomy levels.

Beyond retrieval-based approaches, recent work has explored incorporating structural knowledge into hierarchical classification. [6] leverage knowledge graphs to enhance hierarchical text classification by modeling dependencies between labels, enabling more informed predictions across taxonomy levels. While these methods demonstrate effectiveness in structured classification tasks, they largely depend on static taxonomies and supervised training paradigms. Furthermore, many of these approaches rely on earlier embedding models and loss formulations, limiting their adaptability to evolving taxonomies and large-scale unlabeled datasets. Consequently, their applicability in dynamic educational settings, where both taxonomy structures and question distributions continuously change, remains constrained.

2.2 LLMs as Automated Judges

Recent advances in Large Language Models (LLMs) have led to their adoption as automated evaluators, commonly referred to as *LLM-as-a-Judge*. Despite these advantages, recent work highlights significant limitations of relying solely on LLM-based evaluation. [3] demonstrate that LLM judges are fundamentally constrained by their own task understanding: they tend to perform well only on instances they can solve themselves, leading to unreliable judgments in correctness-critical scenarios. Furthermore, their outputs are susceptible to biases, inconsistencies, and sensitivity to prompt design, particularly in the absence of high-quality human references. While qualitative judging approaches [5] improve interpretability, they still exhibit imperfect alignment with human annotations at the instance level. These findings underscore that LLM-generated scores should be treated as **noisy supervision signals** rather than ground truth, motivating the need for calibration and human grounding before their integration into training pipelines.

2.3 Prediction-Powered Inference and ARES Calibration

Prediction-Powered Inference (PPI) [2] provides a principled statistical framework for combining abundant but noisy machine learning predictions with a small set of high-quality labeled data to obtain valid inference. The core idea is to use the labeled data to estimate and correct the bias in model predictions, enabling the construction of statistically valid confidence intervals without making assumptions about the underlying prediction model. Importantly, PPI guarantees that incorporating predictions does not compromise statistical validity, while potentially improving efficiency by leveraging large unlabeled datasets. Building on this foundation, subsequent work such as PPI++ [1] improves computational and statistical efficiency, adapting the framework to high-dimensional and large-scale settings where predictions vary in quality.

Recent work on automated evaluation systems, such as ARES [7], extends these ideas to the context of LLM-based evaluation. ARES leverages lightweight language model judges to score outputs and employs PPI-based calibration using a small human-annotated dataset to correct systematic biases in these scores. This approach enables reliable evaluation with significantly reduced human supervision, while also providing statistically grounded confidence estimates. By integrating weak supervision from model predictions with limited high-quality annotations, these methods demonstrate a scalable paradigm for transforming noisy evaluation signals into reliable estimates. However, existing approaches primarily focus on aggregate evaluation metrics, leaving open the challenge of adapting such calibration techniques to instance-level decision-making and data selection in downstream learning pipelines.

Chapter 3

Methodology

3.1 Extracting NCERT Taxonomy

The NCERT curriculum for Classes 11 and 12 provides a well-defined hierarchical structure of concepts across Physics and Chemistry. However, this taxonomy is distributed across syllabus PDFs, chapter index pages, and textbook contents, making it non-trivial to directly convert into a machine-readable format. To enable downstream tagging and classification, we performed systematic extraction and hierarchical reconstruction of the NCERT taxonomy as follows.

Level	Description	Count
Subject	Broad discipline (Physics, Chemistry)	2
Chapter	Pedagogical units within each subject	46
Topic	Fine-grained instructional concepts	355

Table: Hierarchical breakdown of the NCERT taxonomy for Class 11–12 Physics and Chemistry.

The detailed composition of this hierarchy is:

- **Physics:** 27 chapters, 210 topics
- **Chemistry:** 19 chapters, 145 topics
- We scraped and parsed the official NCERT syllabus index pages and textbook chapter listings for both subjects.
- The extracted entries contained nested structural information embedded in HTML and PDF formats, often combining chapter titles and sub-topic headings in irregular layouts.
- To reconstruct a clean hierarchical taxonomy, we developed a custom processing pipeline that:
 1. Extracted raw syllabus text blocks for each subject.
 2. Segmented chapter headers and aligned them with their respective sub-topic lists.

3. Normalized variations in formatting (e.g., numbering schemes, indentation, multi-line topic names).
4. Validated parent-child consistency to maintain pedagogical structure:

Subject → Chapter → Topic.

- The final output consists of three structured files (Subject, Chapter, Topic), enabling efficient traversal, hierarchical mapping, and model training.
- Figure 3.1 illustrates a sample decomposition of the NCERT syllabus into a clean three-level hierarchy.

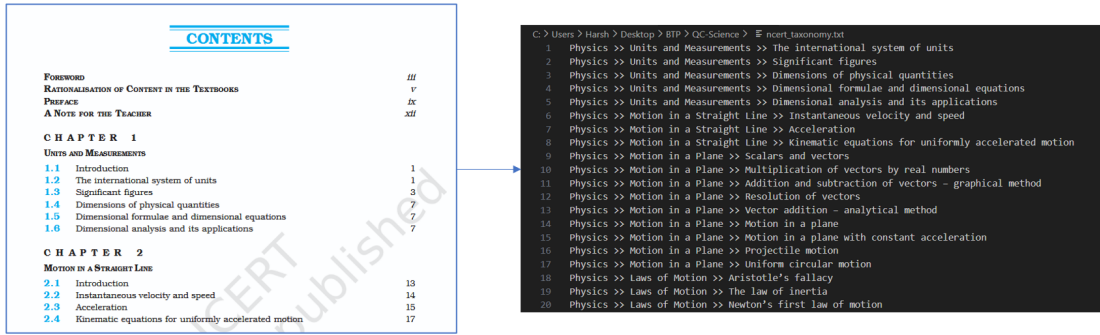


Figure 3.1: Reconstruction of NCERT syllabus into a structured Subject-Chapter-Topic hierarchy

3.2 Weak Supervision via the Taxonomy Classification Model - TaxTag

Building upon the core ideas of TagRec, we introduce **TaxTag**, an updated taxonomy-aware dense retrieval model designed to address limitations of earlier architectures. While TagRec relies on legacy transformer encoders and hinge-based ranking objectives, recent advances in representation learning suggest that contrastive embedding models trained with large-scale weak supervision provide significantly stronger semantic alignment. In particular, modern embedding models such as E5 are trained using contrastive objectives and demonstrate strong transfer performance across retrieval and classification tasks [10]. Motivated by these developments, TaxTag replaces the original encoder with a shared E5-base-v2 backbone and reformulates taxonomy prediction as a dense retrieval problem over label descriptions.

A key design goal of TaxTag is *efficiency without sacrificing performance*. The E5-base-v2 model contains approximately 109M parameters, offering a favorable balance between model size and embedding quality [4]. To further reduce training cost, we employ Low-Rank Adaptation (LoRA), which restricts training to a small set of low-rank parameter updates while keeping the base model frozen. This significantly lowers memory usage and computational overhead compared to full fine-tuning, while preserving generalization capabilities. Empirically, this allows TaxTag to achieve

performance comparable to or better than TagRec-style models, but with reduced training cost and improved scalability.

3.2.1 Model Formulation

Unlike TagRec, TaxTag learns a shared embedding space between questions and taxonomy labels. Also, instead of hinge loss, we adopt an InfoNCE-based contrastive objective, which has been shown to produce more discriminative representations. Each input is prefixed to distinguish between query and label semantics:

$$E(label) = \text{Encoder}(\text{"query:"} + label), \quad (3.1)$$

$$V(question) = \text{Encoder}(\text{"passage:"} + question), \quad (3.2)$$

$$\text{Similarity}(q, t) = \frac{E(q) \cdot V(t)}{\tau}, \quad (3.3)$$

where τ is a temperature parameter controlling the sharpness of the similarity distribution. The use of prefix-based encoding aligns with the training paradigm of E5-style models, which explicitly differentiate query and document representations [10]. This formulation enables TaxTag to generalize across domains and operate in a fully taxonomy-agnostic manner.

To incorporate hierarchical structure, we further extend the contrastive loss with a hierarchy-aware smoothing mechanism. Instead of treating all incorrect labels equally, TaxTag assigns partial credit to semantically nearby labels based on their tree distance. This results in a softer supervision signal that better reflects the graded nature of taxonomy errors. Additionally, hard negative mining is employed to focus learning on the most confusing labels, with penalties scaled according to hierarchical distance. Together, these modifications allow TaxTag to capture both semantic similarity and structural relationships within the taxonomy, leading to more robust and interpretable predictions.

3.2.2 Inference on the Unlabeled Corpus

Our unlabeled dataset $\mathcal{U}$ comprises 27,000 question instances spanning both subjects:

- 15,000 Physics questions
- 12,000 Chemistry questions

For each question $q_i \in \mathcal{U}$, TaxTag computes similarity scores against all taxonomy paths of the form:

$$\text{Subject} \rightarrow \text{Chapter} \rightarrow \text{Topic}.$$

The model returns the top- k most similar taxonomy nodes, and following prior work [8], we set $k = 3$ to retain a diverse yet manageable set of weak labels. For each question q_i , the predicted set is:

$$\mathcal{P}(q_i) = \{(t_1, s_1), (t_2, s_2), (t_3, s_3)\}, \quad (3.4)$$

where t_j denotes a candidate taxonomy label and s_j is its associated cosine similarity score.

3.3 Human Annotation Protocol for Validity Assessment

The weak labels generated by the initial TaxTag model may contain inconsistencies arising from semantic drift, ambiguous question wording, or misalignment between model embeddings and the NCERT taxonomy. To obtain a reliable reference standard for downstream calibration and inference, we designed a structured human annotation protocol to assess the correctness of model-assigned taxonomy labels.

3.3.1 Annotation Guidelines

We developed detailed annotation guidelines to ensure consistent evaluation of question–taxonomy pairs. Annotators were instructed to judge the validity of each predicted taxonomy path at three levels of the hierarchy:

- **Subject Validity** (*Physics* or *Chemistry*)
- **Chapter Validity** (correct mapping to one of the 46 NCERT chapters)
- **Topic Validity** (correct association with one of the 355 fine-grained topics)

Annotation at each level was cast as a binary decision:

$$\text{Valid} = 1, \quad \text{Invalid} = 0,$$

allowing for independent evaluation of errors at different depths of the taxonomy hierarchy. Annotators were asked to consider both semantic relevance and curricular alignment, ensuring that predictions adhered strictly to NCERT’s pedagogical structure.

3.3.2 Creation of the Expert-Labeled Seed Set

To establish a high-quality ground-truth reference for calibration and evaluation, we constructed an expert-labeled seed set from the unlabeled corpus $\mathcal{U}$. We first sampled 600 questions uniformly at random and retrieved their top-3 taxonomy predictions generated by the TagRec model, yielding a total of 1800 question–tag pairs.

Gold Annotation Set. Each pair was independently scored by expert annotators following the standardized validity guidelines. For every question–tag pair (q_i, t_j) , annotators assigned binary

correctness labels at each level of the taxonomy:

$$(q_i, t_j, y_{ij}^S, y_{ij}^C, y_{ij}^T),$$

where y_{ij}^S , y_{ij}^C , and y_{ij}^T denote subject-, chapter-, and topic-level validity respectively (1 = correct, 0 = incorrect). This set of 1800 annotated instances constitutes the *gold reference dataset* used throughout all subsequent supervised analyses.

Evaluation Set. To obtain an additional held-out benchmark for downstream performance assessment, we sampled 500 new questions from the remaining unlabeled pool, again collecting the top-3 predicted taxonomy paths for each. The resulting 1500 question–tag pairs were manually annotated using the same guidelines as the gold set, ensuring consistency and comparability. This evaluation set is later used to measure the end-to-end performance of the proposed taxonomy tagging and calibration pipeline.

Together, the gold reference set and the evaluation set provide the foundational human-labeled data necessary for (i) assessing baseline TagRec behavior, (ii) calibrating LLM-generated validity scores, and (iii) supporting prediction-powered inference for large-scale, label-efficient weak supervision.

3.4 LLM-Based Validity Scoring of Weak Labels

While the expert-labeled seed set provides high-quality ground-truth annotations, manually extending this process to the remaining question–tag pairs is prohibitively expensive. To scale validity assessment across the entire weakly labeled corpus, we employed a large language model (LLM) to automatically estimate the correctness of each predicted taxonomy path.

3.4.1 LLM Judge Design and Prompting Strategy

We used the `openai/gpt-oss-120b` model as an automated annotation judge. The LLM was prompted using a structured, instruction-oriented system message, supplemented with chain-of-thought (CoT) reasoning and few-shot exemplars to ensure consistent judgment across tasks. For each question and its predicted taxonomy path ("Subject → Chapter → Topic"), the model outputs probability estimates for correctness at three hierarchical levels.

$$p^S \in (0, 1), \quad p^C \in (0, 1), \quad p^T \in (0, 1),$$

corresponding respectively to the subject, chapter, and topic Match likelihoods.

The LLM is instructed to provide probabilities with two-decimal precision and to avoid the degenerate endpoints 0.00 and 1.00.

For each unannotated question–tag pair (q_i, t_i) , the LLM judge is queried through a lightweight interface:

$$\text{LLM}(q_i, t_i) \rightarrow \{p_i^S, p_i^C, p_i^T\}, \quad (3.5)$$

where the LLM returns structured JSON output containing level-wise probabilities. Internally, the interface appends the user query:

Question: q_i Predicted Taxonomy: t_i ,

to the pre-defined few-shot prompt template, ensuring identical evaluation conditions across all items.

The exact message prompt used for scoring is shown below.

```
Annotation_message = [
    {
        "role": "system",
        "content": '''
            You are an expert educational content annotator.

            =====
            OBJECTIVE:
            =====

            Evaluate the predicted taxonomy for a given question and assign
            probabilities (0.01{0.99}), 2-decimal precision, and do not include
            endpoints {0.00,1.00} representing the likelihood that:

            1. SUBJECT LEVEL :
               - Probability that the subject semantically matches the main
                 domain/discipline of the question.

            2. CHAPTER LEVEL :
               - Probability that the chapter semantically aligns with the
                 specific subdomain or major topic area involved in the
                 question.

            3. TOPIC LEVEL :
               - Probability that the topic matches the specific concept or
                 knowledge point tested in the question.

            =====
            INPUT:
```

```

=====
1. A natural language question.
2. Taxonomy predicted for the question in the format:
   "Subject >> Chapter >> Topic".

=====
OUTPUT:
=====
Provide probability scores in JSON format.

Example output:
{
    "Subject Level Match": 0.73,
    "Chapter Level Match": 0.26,
    "Topic Level Match": 0.03,
}
'''
},

... (few-shot examples omitted for brevity) ...
]

```

3.4.2 Agreement Between Human and LLM Annotations

To assess the quality of the LLM judge as a proxy for human evaluation, we directly compare the continuous LLM probability scores $p_i^{(\ell)}$ against the binary gold labels $y_i^{(\ell)} \in \{0, 1\}$ using the Mean Absolute Error (MAE):

$$\text{MAE}^{(\ell)} = \frac{1}{n} \sum_{i=1}^n \left| p_i^{(\ell)} - y_i^{(\ell)} \right|, \quad \ell \in \{S, C, T\}.$$

MAE directly captures how far the LLM’s confidence scores deviate from the true binary labels, without requiring an arbitrary binarization threshold. This makes it a natural error metric for the downstream calibration task, where the goal is to bring continuous LLM scores closer to the true validity values.

Table 3.1 summarizes the LLM judge’s raw prediction error at each taxonomy level.

Taxonomy Level	MAE (LLM vs. Gold)
Subject Validity	0.088
Chapter Validity	0.300
Topic Validity	0.344

Table 3.1: Mean Absolute Error between raw LLM probability scores and human gold labels at each taxonomy level.

The low MAE at the subject level (0.088) indicates that the LLM is already highly accurate at judging subject correctness, while the higher MAE values at the chapter (0.300) and topic (0.344) levels reflect the increasing difficulty of fine-grained validity assessment and motivate the need for statistical calibration via prediction-powered inference.

This automatic scoring mechanism provides probabilistic validity assessments for the full dataset, which serve as inputs for subsequent calibration and prediction-powered inference.

3.5 Local Prediction-Powered Inference (L-PPI)

3.5.1 Motivation and Significance

Although the LLM judge provides scalable probabilistic assessments of taxonomy validity, its raw confidence scores are not directly reliable. Prior work has shown that large language models often exhibit systematic overconfidence, distribution shift bias, and poor calibration [angelopoulos2023prediction]. In an educational taxonomy setting, where each question–tag pair must be validated with high precision, such miscalibration can propagate noise into the training pipeline.

Prediction-Powered Inference (PPI) [angelopoulos2023prediction] offers a framework for producing statistically valid uncertainty estimates using a small number of ground-truth labels and a large pool of noisy predictions. However, classical PPI is designed primarily for *aggregate* quantities (e.g., means, proportions) and does not yield calibrated predictions for *individual* instances.

To bridge this gap, we develop **Local Prediction-Powered Inference (L-PPI)**, a local and instance-specific variant of PPI. L-PPI estimates the *conditional bias* and *local uncertainty* of an LLM prediction by examining human-labeled neighbors in an embedding space. This enables:

- **Bias correction:** adjusting raw LLM scores using local residual estimates;
- **Instance-level confidence intervals:** quantifying uncertainty for each question–tag pair;
- **Hierarchical calibration:** computing separate corrections for subject, chapter, and topic predictions.

Using only 300 human-labeled pairs, L-PPI calibrates 27,000 weakly labeled examples, providing a statistically grounded alternative to raw LLM probabilities.

3.5.2 Formal Problem Setup

Let x denote a question–tag pair and $y \in \{0, 1\}$ its human validity label at a given taxonomy level (subject, chapter, or topic). We are given:

- A small, high-quality human-annotated set:

$$\mathcal{L} = \{(x_i, y_i)\}_{i=1}^n,$$

where $n = 300$ in our case.

- An LLM prediction function:

$$f(x) \in [0, 1],$$

providing noisy probability estimates.

- An unlabeled query instance x_0 for which we seek a calibrated, bias-corrected score and a confidence interval.

Define the residual

$$R = f(X) - Y,$$

which quantifies the discrepancy between LLM confidence and human evaluation.

The key identity underlying L-PPI is the PPI decomposition:

$$y_0 = f(x_0) - E[R \mid X \approx x_0] + \text{noise},$$

which states that the true validity at x_0 can be approximated by subtracting the expected local residual from the raw LLM score.

3.5.3 Prediction-Aware Embedding and Local Neighborhood Construction

To approximate the conditional expectation $E[R \mid X \approx x_0]$, L-PPI performs calibration in a *local* region around the query instance x_0 . Locality is defined in a *prediction-aware* embedding space induced by the `all-mpnet-base-v2` sentence encoder.

Rather than embedding only the question text, L-PPI constructs a joint representation that concatenates the question with its predicted taxonomy label:

$$e_i = E(\text{question}_i \text{ [SEP] predicted_taxonomy}_i).$$

This prediction-aware encoding ensures that two questions with identical text but different predicted labels occupy distinct positions in embedding space, enabling neighborhood construction that reflects both semantic content and prediction context.

Let e_0 denote the embedding of the query instance and $d_i = \text{cos_dist}(e_0, e_i)$ the cosine distance to each gold-labeled point. We select the k nearest neighbors:

$$\mathcal{N}_k(x_0) = \text{kNN}(e_0, \{e_i\}, k), \quad \text{with associated distances } \{d_i\}_{i \in \mathcal{N}_k}.$$

For each neighbor $i \in \mathcal{N}_k(x_0)$, we compute the local residual between the LLM score and the human label:

$$r_i = f(x_i) - y_i.$$

3.5.4 Weighted Local Residual Estimation

A key challenge with unweighted residual averaging is *class-imbalance bias*: when the gold-label distribution within the calibration set is skewed (e.g., most neighbors have $y = 1$), the mean residual $\hat{\Delta}$ becomes dominated by the majority class and pushes all corrections in a single direction. This yields strong rectification for the majority class but degrades predictions for the minority class.

To address this, L-PPI employs a *doubly-weighted* residual estimation scheme that assigns higher importance to neighbors that are (a) closer in embedding space and (b) more similar in LLM score to the test point.

Distance weights. Neighbors closer in embedding space contribute more:

$$w_{\text{dist},i} = \frac{1}{d_i + \epsilon}, \quad \epsilon = 10^{-8}.$$

Score-proximity weights. A Gaussian kernel on LLM score difference ensures that the correction is driven by neighbors whose LLM behavior matches the test point, naturally separating high-score and low-score regimes:

$$w_{\text{score},i} = \exp\left(-\frac{|\bar{f}(x_0) - f(x_i)|}{h}\right),$$

where h is a bandwidth hyperparameter controlling the sharpness of score-based weighting.

Combined weights. The final weight for each neighbor is:

$$w_i = \frac{w_{\text{dist},i} \cdot w_{\text{score},i}}{\sum_{j \in \mathcal{N}_k} w_{\text{dist},j} \cdot w_{\text{score},j}}.$$

The weighted local bias estimate is:

$$\hat{\Delta}(x_0) = \sum_{i \in \mathcal{N}_k(x_0)} w_i r_i,$$

and the weighted residual variance is:

$$s_r^2(x_0) = \frac{\sum_{i \in \mathcal{N}_k} w_i (r_i - \hat{\Delta})^2}{1 - \sum_{i \in \mathcal{N}_k} w_i^2},$$

following the standard reliability-weights formula for unbiased variance estimation.

This formulation ensures that the bias correction $\hat{\Delta}$ is driven primarily by neighbors that are both semantically similar and exhibit comparable LLM scoring behavior, mitigating the class-imbalance bias inherent in unweighted averaging.

3.5.5 LLM Sampling Variability

Because LLM predictions may vary slightly across repeated queries, L-PPI performs m independent evaluations:

$$f_j(x_0), \quad j = 1, \dots, m.$$

Define:

$$\bar{f}(x_0) = \frac{1}{m} \sum_{j=1}^m f_j(x_0), \quad s_m^2(x_0) = \frac{1}{m-1} \sum_{j=1}^m (f_j(x_0) - \bar{f}(x_0))^2.$$

3.5.6 Bias-Corrected Score with Confidence-Based Shrinkage

The combined standard error is:

$$SE(x_0) = \sqrt{\frac{s_r^2(x_0)}{k} + \frac{s_m^2(x_0)}{m}}.$$

When the neighborhood residuals are highly variable (large SE), the estimated bias $\hat{\Delta}$ may be unreliable. To prevent noisy corrections from degrading predictions, L-PPI applies a *shrinkage damping* factor that attenuates the correction proportionally to the standard error:

$$\gamma(x_0) = \frac{1}{1 + \lambda \cdot SE(x_0)},$$

where $\lambda > 0$ is a shrinkage hyperparameter. When the neighborhood is consistent ($SE \approx 0$), $\gamma \approx 1$ and the full correction is applied. When the neighborhood is noisy (SE is large), $\gamma \rightarrow 0$ and the method conservatively retains the raw LLM score.

The bias-corrected prediction is:

$$\hat{y}_{\text{rect}}(x_0) = \bar{f}(x_0) - \gamma(x_0) \cdot \hat{\Delta}(x_0).$$

3.5.7 Degrees of Freedom

Following the Welch–Satterthwaite approximation, the effective degrees of freedom are:

$$V = \frac{s_r^2(x_0)}{k} + \frac{s_m^2(x_0)}{m},$$
$$df = \frac{V^2}{\frac{(s_r^2(x_0))^2}{(k-1)k^2} + \frac{(s_m^2(x_0))^2}{(m-1)m^2}}.$$

3.5.8 Confidence Interval Construction

Let

$$z = \begin{cases} \Phi^{-1}(1 - \alpha/2), & df > 30, \\ t_{df}^{-1}(1 - \alpha/2), & df \leq 30. \end{cases}$$

The instance-level calibrated confidence interval is:

$$CI(x_0) = [\hat{y}_{\text{rect}}(x_0) - z \cdot SE(x_0), \hat{y}_{\text{rect}}(x_0) + z \cdot SE(x_0)].$$

L-PPI Algorithm (clean pseudocode)

Algorithm 1 Local Prediction-Powered Inference (L-PPI)

Input :

- Query instance x_0 (question–tag pair)
- Human-labeled calibration set $\mathcal{L} = \{(x_i, y_i)\}_{i=1}^n$
- LLM scoring function $f(\cdot)$
- Embedding encoder $E(\cdot)$ (prediction-aware)
- Neighbors k , LLM repeats m , bandwidth h , shrinkage λ , confidence α

Output: Bias-corrected score $\hat{y}_{\text{rect}}$, standard error SE , confidence interval CI

1. Prediction-aware embeddings

$e_0 \leftarrow E(\text{question}_0 [\text{SEP}] \text{taxonomy}_0)$ $e_i \leftarrow E(\text{question}_i [\text{SEP}] \text{taxonomy}_i)$ for all $(x_i, y_i) \in \mathcal{L}$

2. Select local neighborhood with distances

$\mathcal{N}, \{d_i\} \leftarrow \text{kNN}(e_0, \{e_i\}, k)$

3. LLM consensus at x_0

for $j \leftarrow 1$ to m do

$f_j \leftarrow f(x_0)$

end

$\bar{f} \leftarrow \frac{1}{m} \sum_j f_j$, $s_m^2 \leftarrow \frac{1}{m-1} \sum_j (f_j - \bar{f})^2$

4. Compute weighted residuals

foreach $i \in \mathcal{N}$ do

$r_i \leftarrow f(x_i) - y_i$ $w_{\text{dist},i} \leftarrow 1/(d_i + \epsilon)$ $w_{\text{score},i} \leftarrow \exp(-|\bar{f} - f(x_i)|/h)$ $w_i \leftarrow w_{\text{dist},i} \cdot w_{\text{score},i}$

end

Normalize: $w_i \leftarrow w_i / \sum_j w_j$ $\hat{\Delta} \leftarrow \sum_i w_i r_i$, $s_r^2 \leftarrow \frac{\sum_i w_i (r_i - \hat{\Delta})^2}{1 - \sum_i w_i^2}$

5. Combined SE and shrinkage

$SE \leftarrow \sqrt{s_r^2/k + s_m^2/m}$ $\gamma \leftarrow 1/(1 + \lambda \cdot SE)$ $\hat{y}_{\text{rect}} \leftarrow \bar{f} - \gamma \cdot \hat{\Delta}$

6. Degrees of freedom (Welch--Satterthwaite)

$V \leftarrow s_r^2/k + s_m^2/m$ $df \leftarrow V^2 / \left(\frac{(s_r^2)^2}{(k-1)k^2} + \frac{(s_m^2)^2}{(m-1)m^2} \right)$

7. Critical value and CI

if $df > 30$ then

$z \leftarrow \Phi^{-1}(1 - \alpha/2)$

end

else

$z \leftarrow t_{df}^{-1}(1 - \alpha/2)$

end

$CI \leftarrow [\hat{y}_{\text{rect}} - z \cdot SE, \hat{y}_{\text{rect}} + z \cdot SE]$

return $\hat{y}_{\text{rect}}, SE, CI$

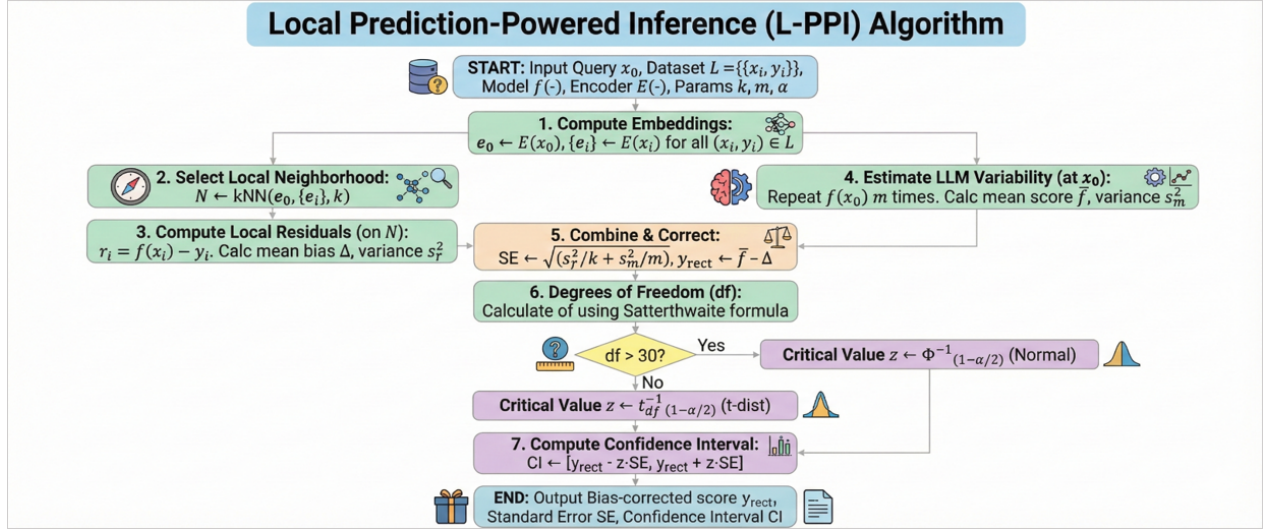


Figure 3.2: L-PPI Algorithm flowchart

3.6 Iterative Training via Rectified-Score Filtering

With L-PPI providing calibrated validity scores and confidence intervals for all LLM-judged question–tag pairs, the final stage of the pipeline uses these rectified assessments to construct high-quality training data for iteratively improving the TaxTag taxonomy classifier.

3.6.1 Motivation

The TaxTag model produces initial weak taxonomy predictions, but these predictions contain errors that propagate if used directly for retraining. The LLM judge provides probabilistic validity assessments, but as demonstrated by the MAE analysis, these raw scores carry systematic biases that vary across taxonomy levels. L-PPI rectifies these biases and additionally provides confidence intervals that quantify the reliability of each corrected score. This enables a principled filtering strategy: retain only those question–tag pairs for which the rectified score indicates high validity *and* the confidence interval is sufficiently narrow.

3.6.2 Confidence-Based Gold Label Selection

For each question–tag pair (q_i, t_j) scored by the LLM and rectified by L-PPI, we obtain:

- Rectified validity scores: $\hat{y}_{\text{rect}}^{(S)}, \hat{y}_{\text{rect}}^{(C)}, \hat{y}_{\text{rect}}^{(T)}$
- Confidence interval half-widths: $CI^{(S)}, CI^{(C)}, CI^{(T)}$

A question–tag pair is selected as a high-confidence training example if it satisfies all of the following conditions:

1. The rectified topic-level score exceeds a validity threshold τ :

$$\hat{y}_{\text{rect}}^{(T)}(q_i, t_j) \geq \tau.$$

2. The confidence interval at the topic level is narrow (high certainty):

$$CI^{(T)}(q_i, t_j) \leq \delta.$$

The filtered set of high-confidence pairs is:

$$\mathcal{D}_{\text{filtered}} = \left\{ (q_i, t_j) : \hat{y}_{\text{rect}}^{(T)} \geq \tau \text{ and } CI^{(T)} \leq \delta \right\}.$$

This set serves as pseudo-gold training data for the next round of TaxTag fine-tuning.

3.6.3 Iterative Retraining Loop

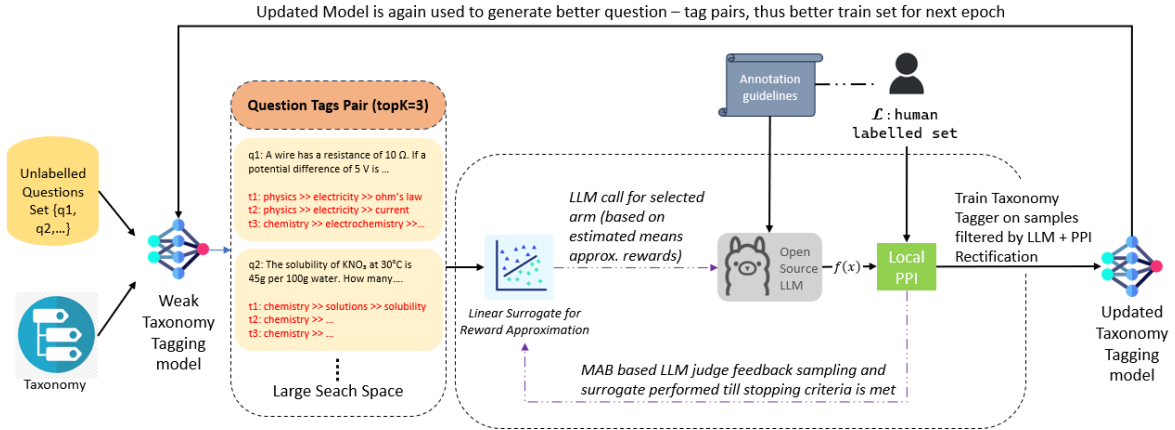


Figure 3.3: Overview of the iterative training framework. L-PPI calibrates LLM-generated validity scores, which are then filtered by confidence to produce high-quality pseudo-gold labels for incrementally retraining the TaxTag taxonomy classifier.

The overall training procedure operates as a closed loop across multiple epochs:

1. **Predict:** TaxTag generates top- k taxonomy predictions for all questions in the unlabeled corpus $\mathcal{U}$.
2. **Score:** The LLM judge assigns validity probabilities to each question-tag pair.
3. **Rectify:** L-PPI calibrates the LLM scores using the human-labeled gold set, producing rectified scores and confidence intervals.
4. **Filter:** High-confidence pairs are selected from the rectified output using the threshold criteria above.

5. **Retrain:** TaxTag is fine-tuned on $\mathcal{D}_{\text{filtered}}$, using the selected taxonomy labels as training targets.

At each subsequent iteration, the improved TaxTag model produces better predictions, which in turn lead to higher LLM scores, tighter confidence intervals, and a larger pool of high-quality filtered training data. This positive feedback loop progressively expands the effective training set while maintaining label quality.

In summary, the iterative training framework leverages L-PPI’s calibrated uncertainty estimates as a principled gatekeeper for training data selection, enabling the taxonomy classifier to improve over multiple rounds without requiring additional human annotation beyond the initial seed set.

Chapter 4

Results

4.1 Local PPI Calibration Analysis

To evaluate the effectiveness of our LLM-based scoring and Local PPI calibration framework, we analyzed rectified predictions on the 51 manually annotated question–tag pairs of the held-out evaluation set. At each taxonomy level (subject, chapter, topic), we compared:

- the human-provided binary validity labels (blue),
- the raw LLM probability estimates (orange), and
- the rectified probabilities produced by L-PPI (green),

with neighborhood-based confidence intervals indicated by vertical grey bars. Figures 4.1, 4.2, and 4.3 visualize these comparisons across all instances.

Across all levels, three consistent qualitative behaviors emerge:

- **Correct predictions (manual=1):** The rectified probabilities are typically *higher* than the raw LLM scores, pushing predictions closer to 1. This manifests in the plots as the green points lying *above* the orange points.
- **Incorrect predictions (manual=0):** The rectifier reduces LLM overconfidence by pulling the probability downward toward 0. Visually, green points appear *below* their corresponding orange points.
- **Confidence intervals (CIs):**
 - Correct rectifications tend to have *tight* CIs (high neighborhood agreement).
 - Incorrect rectifications tend to have *wide* CIs (low neighborhood agreement).
 - Points that remain incorrect even after rectification but with narrow CIs can be *filtered out*, improving dataset quality.

4.2 Quantitative Results

Table 4.1 summarizes rectification accuracy and confidence-interval statistics across all levels. A rectification is counted as *correct* when the rectified probability moves in the direction of the human label (increasing toward 1 for manual=1, decreasing toward 0 for manual=0).

Metric	Subject	Chapter	Topic
Overall Right Rectification (%)	0.549	0.490	0.647
Avg CI Size (Correct)	0.065	0.338	0.344
Avg CI Size (Incorrect)	0.400	0.466	0.665
Manual=0			
Pct Right Rectification	0.375	0.741	0.714
Avg CI Size (Manual=0)	0.149	0.340	0.449
Manual=1			
Pct Right Rectification	0.581	0.208	0.375
Avg CI Size (Manual=1)	0.229	0.475	0.486

Table 4.1: Summary of rectification accuracy and confidence interval statistics across taxonomy levels.

Several noteworthy patterns emerge:

- Topic-level rectification performs the strongest overall (64.7%), reflecting the usefulness of local residual calibration for fine-grained distinctions.
- Manual=0 cases show particularly high correction accuracy for Chapter (74.1%) and Topic (71.4%), demonstrating L-PPI’s ability to dampen false-positive confidence.
- CI sizes behave as expected: incorrect rectifications are consistently associated with *larger* intervals.

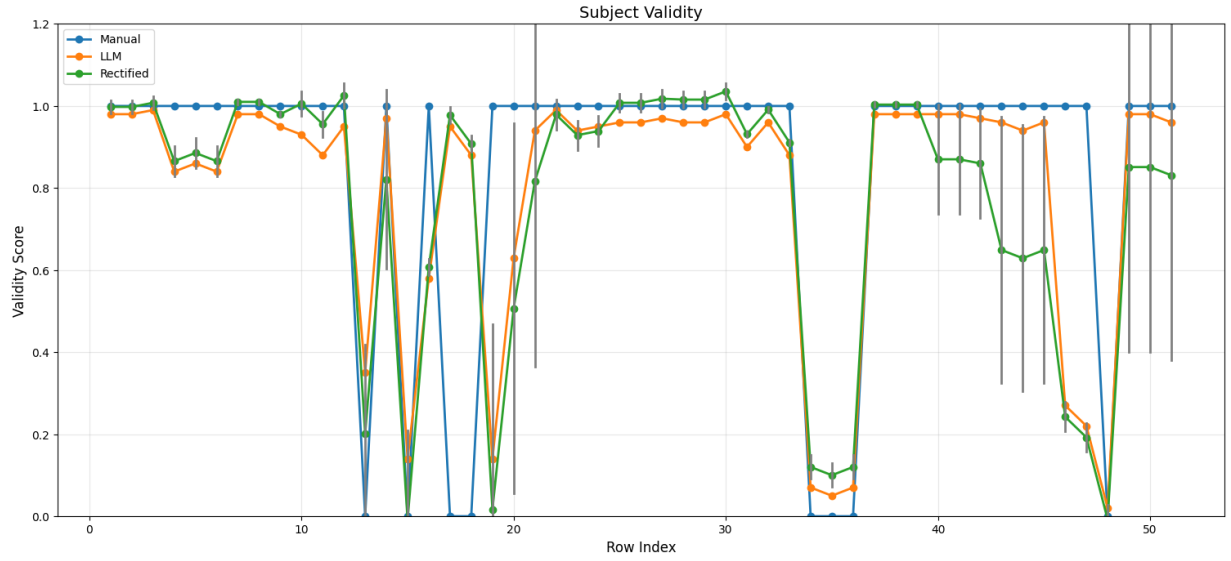


Figure 4.1: Subject-Level Validity: Manual (blue), LLM (orange), and Rectified (green) with confidence intervals.

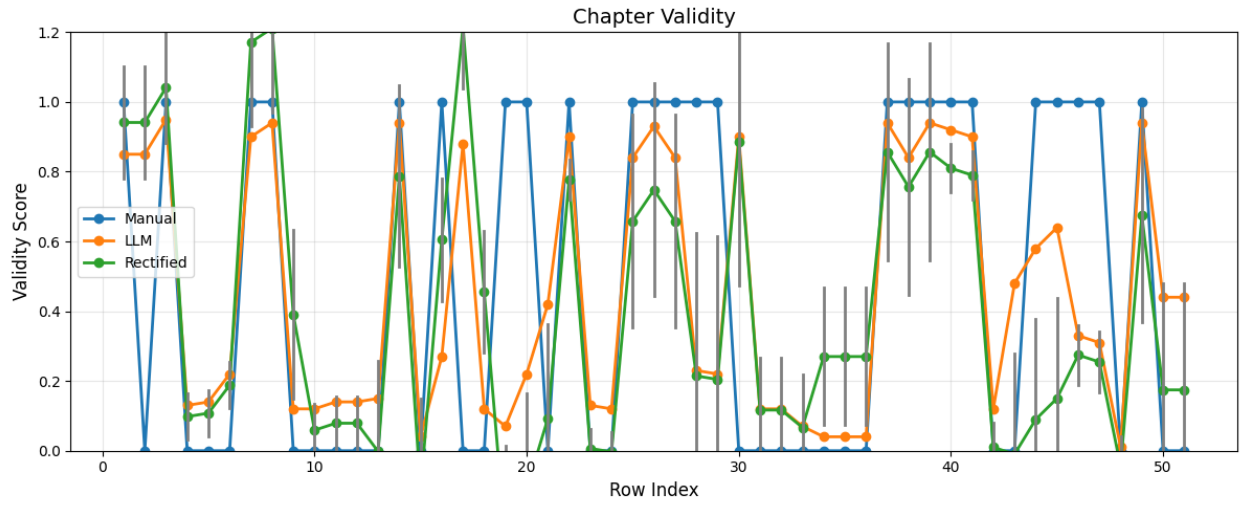


Figure 4.2: Chapter-Level Validity: Manual (blue), LLM (orange), and Rectified (green) with confidence intervals.

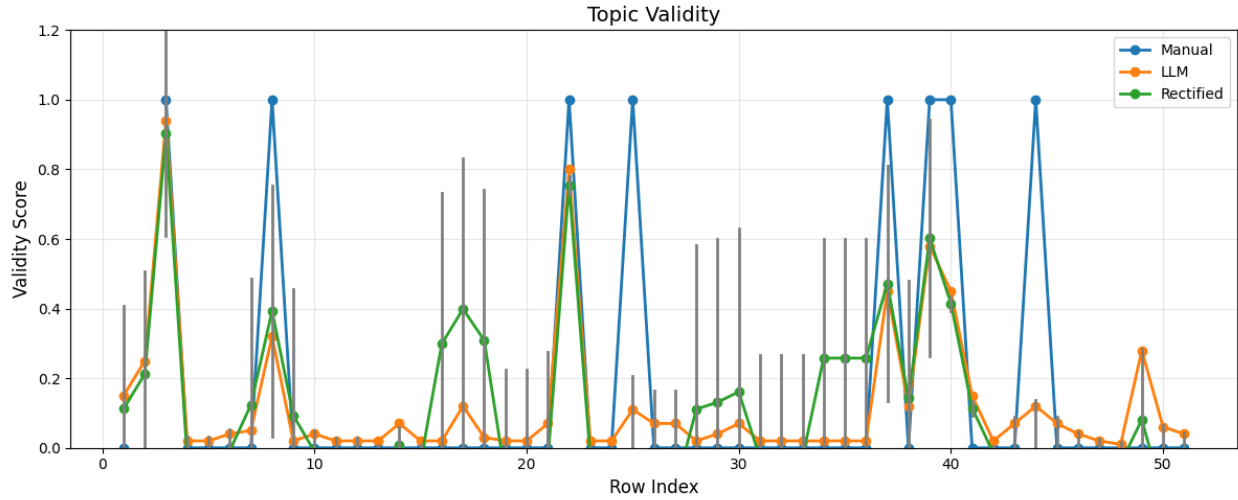


Figure 4.3: Topic-Level Validity: Manual (blue), LLM (orange), and Rectified (green) with confidence intervals.

4.3 Case Studies

Subject-Level Example. Table 4.2 shows a representative example where the LLM is already confident, and L-PPI appropriately increases the score while producing a very tight CI.

Subject-Level Analysis				
Question: (Optics problem omitted here for brevity; identical to Section text.)				
Tag: Physics → Ray Optics and Optical Instruments → Optical Instruments				
Manual: 1 LLM: 0.88 Rectified: 0.9978 CI: [0.9825, 1.0130]				
Interpretation: Strong upward correction with tight CI, reflecting high local consistency.				

Table 4.2: Subject-level rectification example.

Topic-Level Example. Table 4.3 shows a case where the predicted topic is incorrect. The rectifier correctly pushes the score downward and produces a wide CI, signaling uncertainty.

Topic-Level Analysis				
Question: (Optics problem omitted for brevity.)				
Tag: Physics → Ray Optics and Optical Instruments → Optical Instruments				
Manual: 0 LLM: 0.15 Rectified: 0.1122 CI: [−0.1820, 0.4064]				
Interpretation: Downward correction with wide CI, reflecting neighbor disagreement at fine-grained level.				

Table 4.3: Topic-level rectification example.

4.4 Effect of CI-Based Filtering

A central benefit of L-PPI is that confidence intervals can be used to filter out unreliable predictions. We apply thresholding based on CI width at each taxonomy level:

Subject: CI width < 0.40 , Chapter: CI width < 0.45 , Topic: CI width < 0.60 .

The resulting accuracy improvements are shown in Table 4.4.

Level	Right Rectification After Filtering
Subject	0.667
Chapter	0.593
Topic	0.818

Table 4.4: Rectification accuracy after CI-based filtering.

Key Observation. Across all levels, CI filtering produces a substantial increase in correctness:

- Subject: from 0.549 $\rightarrow$ 0.667
- Chapter: from 0.490 $\rightarrow$ 0.593
- Topic: from 0.647 $\rightarrow$ 0.818

This demonstrates that CI-based filtering is an effective mechanism for isolating high-reliability instances, which is essential for constructing the next iteration of the taxonomy tagger training set.

4.5 Summary

Local PPI provides consistent upward or downward corrections to LLM scores in alignment with human annotations and equips each prediction with a principled uncertainty interval. CI-based filtering significantly improves the reliability of the rectified predictions, making the resulting high-confidence subset well-suited for downstream retraining of the taxonomy classification model.

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